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香 港 中 文 大 學
The Chinese University of Hong Kong

二零二五至二六年度上學期科目考試 (期中)

Course Examination 1st Term, 2025-26

科目編號及名稱
Course Code & Title: CSCI4430 Data Communications and Computer Networks

時間
Time allowed: 0 小時 hours 25 分鐘 minutes

學號
Student I.D. No. ______________ 姓名 Name ______________

This is a **closed-book** exam. One approved calculator is allowed.

There are **3** questions. Answer **ALL** questions in the space provided on the exam paper.

Write down your **student ID** and **name**.

Question	1	2	3	Total
Points	20	20	10	50
Score				

1. (20 points) **Single Choice Questions**

(1) (2 points) Which of the following statements correctly describes the difference between circuit switching and packet switching?

A. Circuit switching reserves network resources per connection, while packet switching treats each packet independently without prior reservation.

B. Packet switching has lower transmission delay than circuit switching because it does not require circuit setup.

C. Circuit switching is more efficient for bursty traffic, while packet switching is better for constant-bit-rate traffic.

D. Packet switching uses a dedicated path for each connection, while circuit switching shares paths among multiple packets.

Answer: A

(2) (2 points) Which of the following is INCORRECT about performance metrics of a network?

A. For two server nodes directly connected via a link, the end-to-end delay between them equals the sum of transmission delay and propagation delay.

B. For two server nodes connected to a switch, the end-to-end delay between them equals the sum of transmission delay, propagation delay and processing delay.

C. Queueing delay is only related to arrival rate at the queue and transmission rate of the outgoing link.

D. Persistent overload at a switch will lead to packet drop.

Answer: B or C

(3) (2 points) Which of the following is INCORRECT about the layers in network stack?

A. HTTP is an application-layer protocol.

B. Transport layer should be implemented by both end hosts and the network.

C. Network layer (L3) routes packets between networks for global delivery.

D. Layering reduces complexity of the network but also leads to overhead.

Answer: B

(4) (2 points) Which of the following is a key characteristic of the "narrow waist" of the Internet protocol stack (centered on IP)?

A. IP must be modified whenever new application-layer protocols (e.g., HTTP/3) are introduced.

B. All application-layer protocols must directly interact with the data link layer to ensure compatibility.

C. IP decouples upper-layer applications from lower-layer network technologies, enabling independent evolution of both.

D. The data link layer and physical layer depend on IP to define their frame formats.

Answer: C

(5) (2 points) When a web browser sends an HTTP request, the data is encapsulated as it moves down the protocol stack. At the Transport Layer, what is the payload of a TCP segment?

A. The IP packet

B. The Ethernet frame

C. The HTTP request message

D. The physical bits

Answer: C

(6) (2 points) What is the primary purpose of the Domain Name System (DNS)?
A. To encrypt web traffic between clients and servers.
B. To map human-readable domain names to IP addresses.
C. To store web page content for faster access.
D. To manage email delivery between mail servers.
Answer: B

(7) (2 points) What is the primary function of a reverse proxy cache in the context of a Content Delivery Network (CDN)?
A. To reduce latency for clients by caching frequently accessed content close to them, typically deployed by ISPs.
B. To reduce the processing and bandwidth load on origin servers by caching content close to the servers themselves.
C. To translate private IP addresses used within a content provider's network to public IP addresses.
D. To encrypt all traffic between clients and the origin server for enhanced security.
Answer: B

(8) (2 points) What is the primary motivation for using Clos or fat-tree topologies in modern data center networks?
A. To minimize the physical number of switches and cables required, thus reducing capital expenditure.
B. To provide high bisection bandwidth, allowing communication-intensive applications to run without internal network bottlenecks.
C. To simplify routing by ensuring there is only a single, fixed path between any two servers.
D. To prioritize traffic going out to the public internet (North-South traffic) over internal server-to-server traffic (East-West traffic).
Answer: B

(9) (2 points) In TCP socket programming, which sequence of system calls is typically used by a server to prepare for and accept a new client connection?
A. socket() -> connect() -> accept() -> recv()
B. socket() -> bind() -> listen() -> accept()
C. socket() -> listen() -> bind() -> send()
D. socket() -> accept() -> listen() -> bind()
Answer: B

(10) (2 points) In the context of reliable transport protocols, what is the core advantage of Selective Repeat (SR) over Go-Back-N (GBN)?
A. SR is only suitable for small window sizes, while GBN works for large windows.
B. SR only retransmits lost packets, while GBN retransmits all unacknowledged packets after a timeout, saving bandwidth in high-error networks.
C. SR does not need timers, eliminating timer synchronization issues.
D. SR uses cumulative acknowledgments, which are simpler to implement than GBN's selective acknowledgments.
Answer: B

2. (20 points) **Short Questions**

(1) (10 points) How can we improve the efficiency of HTTP? Briefly explain one approach.

Answer: Persistent connections; pipelining; parallel connections; any other valid thoughts are fine

Notes: Caching or CDN is acceptable

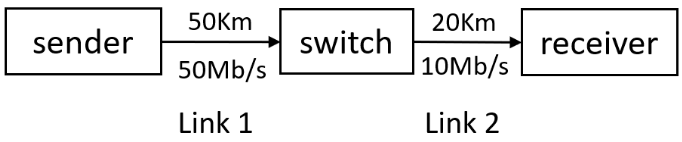
(2) (10 points) Explain the role of the accept() system call in TCP socket programming for a server. Why does it create a new file descriptor instead of reusing the listening socket?

Answer:

- After listen(): The accept() system call is used by a TCP server to handle incoming client connections. It is called after the server calls listen() to put the listening socket in a "passive mode" (waiting for connections).
- Block/wait: accept() blocks until a client initiates a connection via connect().
- Create a socket: When a connection is established, accept() creates a new file descriptor (socket) dedicated to communication with that specific client.
- Different function: It does not reuse the listening socket because the listening socket is reserved for accepting new connection requests (it remains in passive mode to handle other clients).
- New identifier: Using a new socket per client allows the server to distinguish between multiple concurrent connections: each new socket maps to a unique 4-tuple (server IP, server port, client IP, client port), enabling the server to send/receive data to/from each client independently (e.g., via read()/write() on the new socket). For example, a web server can use the listening socket (port 80) to accept connections from 100 clients, and accept() will create 100 new sockets — one for each client to handle HTTP requests.

3. (10 points) **Performance Analysis**

Consider the figure below. The sender connects to the receiver through one store-and-forward switch. The sender generates a stream of 1000-byte data packets to the receiver and is assumed to always have data to send. The propagation speed in all links is 2.5 x 10^8 m/s. The sender, switch, and receiver have negligible processing and queuing delay, and the size of ACK packets is also considered negligible. (Note: 1 Byte = 8 bits)



(a) What is the average throughput of this connection in Mb/s?

Answer: d1 = 0.2ms, d2=0.08ms, d_total = d1+d2 = 0.28ms

Packet size in bits: 1000bytes * 8=8000bits; t1=0.16ms, t2=0.8ms; t_cycle = t1+t2+RTT = 1.52ms; throughput = 8000/(1.52*10^-3s) = 5.26 Mb/s

(b) Based on the throughput calculated in part (a), what is the utilization of Link 1 and Link 2?

Answer: 5.26/50 = 10.52%; 5.26/10 = 52.6%